

# Credit Card Fraud Detection Analysis Report

## Executive Summary

This project focuses on detecting fraudulent financial transactions using machine learning. The analysis includes data preprocessing, exploratory data analysis, feature engineering, model development, and evaluation using Logistic Regression.

## Problem Statement

Financial fraud causes significant losses for financial institutions. The objective of this project is to analyze transaction data and build a classification model capable of distinguishing fraudulent transactions from legitimate ones.

## Dataset Overview

The dataset consists of transaction records containing information such as transaction type, amount, account balances, and fraud labels.

Key variables include:

- Transaction Type
- Amount
- Origin Balance
- Destination Balance
- Fraud Indicator
- Flagged Fraud Indicator

## Methodology

### Data Exploration

- Examined dataset structure.
- Checked missing values.
- Studied fraud distribution.
- Explored transaction categories.

### Data Preprocessing

- Removed unnecessary columns.
- Applied feature scaling.
- Performed one-hot encoding for categorical variables.

### Feature Engineering

Generated new balance-based features:

- balanceDifforig

- balanceDiffdest

## Exploratory Data Analysis

The following analyses were performed:

- Transaction type distribution
- Fraud distribution
- Transaction amount distribution
- Fraud by transaction type
- Correlation analysis
- Sender and receiver analysis
- Fraud analysis for TRANSFER and CASH\_OUT transactions

## Machine Learning

Algorithm Used:

- Logistic Regression with balanced class weights

Evaluation Metrics:

- Classification Report
- Confusion Matrix
- Accuracy Score

## Key Findings

- Fraudulent transactions represent a small portion of the overall dataset.
- Fraud occurrence varies significantly across transaction types.
- Balance-based engineered features provide additional predictive information.
- Feature scaling and categorical encoding improve model preparation.

## Business Impact

A fraud detection model can assist financial institutions by:

- Identifying suspicious transactions.
- Supporting fraud investigation.
- Reducing financial losses.
- Improving transaction monitoring.

## Conclusion

The project demonstrates a complete machine learning workflow for fraud detection, from preprocessing and exploratory analysis to model development and evaluation. The Logistic Regression model provides a baseline solution for identifying fraudulent financial transactions and establishes a foundation for future model improvements.